JSynth

The flocking synth. User Manual



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0.10.4	CLAP · Instrument · MPE	macOS · Windows · Linux	October 2026

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About this manual

Chapters 2 and 3 get you installed and playing in a few minutes. Chapter 4 tours the panel, and Chapter 5 explains how a note becomes a sound, which is the quickest way to understand what every control does. Chapters 6 to 13 cover the sky, the parts and every panel. Chapter 14 lists the factory presets and Chapter

15 has recipes to start from. The appendix lists every parameter with its exact range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **TUNE PULL**. Every control is a host parameter, and every one that moves continuously can be modulated by the host. A tag beside a parameter means:

ROUTE A route can move it, per note, and the host can modulate it per note too.

CHAPTER 01

Welcome to JSynth

A two-part MPE synth where every note is a creature, and the creatures flock.

Play a note on JSynth and a creature hatches in the sky in the middle of the panel. Your note is its **lure**: low notes pull their creatures to the left, high notes to the right, and sliding your finger up an MPE controller lifts it higher. Press harder and the creature flies faster and burns brighter. Let go and it flies off as its sound fades, trailing ink.

While they are held, the creatures do what flocks do. They gather, turn together, keep a little apart, and wander. And because each creature is a voice, how it flies can become how it sounds: how fast it goes, how near its nearest neighbour is, which way it heads and how hard it turns are all modulation sources, as exact as an envelope. Creatures can also hear each other directly. A neighbour's oscillator can frequency-modulate this one, creatures flying the same way drift toward pure intervals, and a hunter that catches its prey rings it.

Underneath is a serious synth. Each of the two parts, **A** and **B**, is its own species with its own patch: three oscillators with analog and FM waves, a 3×3 FM grid, noise, drive, two filters, three envelopes, four LFOs, four mod buses and sixty-four routes. Nothing the flock does reaches the sound unless you route it, so a plain patch is plain.

At a glance

- **Two parts**, layered, split, by velocity, by MPE zone, or crossfaded by expression. Up to 16 voices each.
- **Three oscillators** per voice: sine, triangle, saw, square, pulse, the TX81Z's FM waves and noise, pitched by note, ratio or fixed Hz.
- **A 3×3 FM grid**, phase or exponential, whose diagonal is each oscillator's feedback.
- **Two filters** per voice: SVF up to 48 dB, notch, peak, ladder and comb, in serial or parallel, with each oscillator picking its filter.
- **MPE** and CLAP note expressions: velocity, press, glide, slide and release velocity, each with its own curve and smoothing.
- **The flock**: boids, each creature following its neighbours by cohesion, alignment and separation, plus speed, wander, lure and range, with four behaviours: Shoal, Scatter, Hunt and Flee.
- **Four flock sources** per creature: Speed, Nearest, Heading and Turn.
- **Between creatures**: Neighbour FM, Tune Pull and Bite.
- **Shape envelopes** after XAOC's Zadar, 60 contours from 1 ms to 30 minutes, and **four mod buses** per part.
- **MAP**: touch a source, drag a knob. No matrix page, and up to 64 routes per part.
- **166 factory presets**, mostly showcases: the flock playing the sound, Press shaped by the mod buses, drones and textures on Shape envelopes, and patches that move on their own or move each other, with a few plain patches to start from.

> **THE FLOCK IS THE ENGINE'S**
> The flock flies on the audio thread, at a fixed rate, inside the synth. What you see in the sky is exactly what the voices hear, and a render plays back the same every time from the same start.

CHAPTER 02

Installation

JSynth is a CLAP plug-in. It ships in the JClap bundle alongside the other JClap plug-ins.

Requirements

Plug-in format	CLAP 1.2 (instrument, stereo). Your host must support CLAP, for example Bitwig Studio, REAPER, FL Studio or Studio One.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11.
Controller	Any MIDI keyboard. An MPE controller (a Seaboard, Linnstrument, Osmose or similar) or a host that sends CLAP note expressions brings out the most.

Installing

01 **Download**

Grab the zip for your operating system from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. It contains every JClap plug-in.

02 **Copy the plug-in**

Copy `JSynth.clap` (a bundle on macOS) into your CLAP folder:

macOS	`~/Library/Audio/Plug-Ins/CLAP`
Windows	`C:\Program Files\Common Files\CLAP`
Linux	`~/.clap`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. JSynth appears as an instrument.

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. The simplest install is from Terminal, since files fetched with `curl` are not flagged:

```
curl -sL https://github.com/jeffmhopkins/Jclap/releases/latest/download/jclap-macOS.zip -o /tmp/jclap.zip
&& mkdir -p ~/Library/Audio/Plug-Ins/CLAP && ditto -x -k /tmp/jclap.zip ~/Library/Audio/Plug-Ins/CLAP
```

If you already downloaded with a browser, clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap
```

Outputs

JSynth has one note input and three stereo outputs: **Main**, **Part A** and **Part B**. Normally both parts play on Main. Set a part's **Output** to *Own* (a host parameter) and it plays on its own port instead, so you can process the two species separately in your host.

CHAPTER 03

Quick Start

Five minutes from an empty instrument slot to a flock you can hear.

01 **Load JSynth and play**

Put JSynth on an instrument track and play a few notes. Each note hatches a creature at its lure: low notes on the left, high on the right. Hold a chord and watch them gather; let go and they fly off as they fade. No keyboard to hand? Click in the sky: left to right picks the note (C2 to C6), and dragging bends it.

02 **Pick a preset**

Click the preset name in the header to open the menu, grouped by category: *Essentials*, a few plain patches by instrument family, then the showcases: *Flock*, *Pressure*, *Expressive*, *Drones*, *Textures*, *Motion*, *Interplay*, *Landscapes* and *Monsta*. Or click the arrows, or scroll over the bar, to step through them. Try *Tine Piano*, then *Shifting Glass*, then *Solo Violin* with some pressure, then *Predator and Prey* and *Gathering Voices*.

03 **Let the creatures hear each other**

Load *Init* and turn **OSC 2**'s **LEVEL** up to about half, since Neighbour FM plays through Osc 2. Then open the **FM** tab at the top left and turn up **FLOCK FM**. Now hold a chord: when creatures fly close together they frequency-modulate each other, and the sound rings and shifts as the flock moves. Raise **TUNE PULL** too and creatures flying the same way drift into just intonation.

04 **Map a flock source**

Click **SPEED** under the sky, which turns **MAP** on with Speed armed. Then drag Filter 1's **CUTOFF** up on the **FILTER** panel at the left. A ring in Speed's colour shows the route while Speed is armed. Press **MAP** again to finish. Now the faster a creature flies, the brighter it sounds; play harder (Press makes creatures faster) to hear it.

05 **Add a second species**

Set the play mode in the header to **LAYER** and click **PART B**. Part B is a whole second patch, and by default it hunts: its creatures chase part A's. On part B's **FM** tab, raise **BITE**, and every creature B catches rings with its hunter's tone.

06 **Save it**

Click **SAVE** in the preset bar, type a name and press Enter. Your preset appears under *User* in the preset menu. Settings are also saved with your project automatically.

> **TIP**
> Every control has a tooltip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

The sound sits down the left in three big panels that draw what they do, the sky fills the right, and the voice and the output sit under it. Everything edits the part you have chosen.



1. **Preset bar.** Step with the arrows, click the name for the menu, **SAVE** your own.
2. **PART A / PART B.** Which species the panels edit.
3. **Play mode.** Which part each note plays.
4. **MAP.** Mapping: touch a source, drag a knob. Touching a source turns it on too.
5. **Touch chips.** VELOCITY, PRESS, GLIDE, SLIDE, RELEASE: sources that light as you play.
6. **OSCILLATORS / FM.** The three oscillators with their waves drawn; the FM grid, the links between creatures and the noise.
7. **FILTER.** Both filters' response, drawn live, and their controls.
8. **ENVELOPES / LFOS / BUSES.** ENV 1 AMP, ENV 2 and ENV 3; LFO 1 and 2; the four mod buses.
9. **The sky.** Both species, their ink, and the links between them. Click to play. Over its top left, the look (SUMI or AURORA) and the palette.
10. **Flock sources.** The creatures' own modulation sources, and Random.

11 **RINGS, ROUTES.** Lure rings and the route list.

12 **VOICE / FLOCK / MOTION / TOUCH.** The voice, how the species flies, the wingbeat and the wind, and how each touch responds.

13 **Meters.** The levels of A, B and the output as they play, with each one's setting marked.

14 **SYNTH / SKY.** The two views: the panels beside the sky, or the sky alone.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
`Shift` **+ drag**	Fine adjustment.
Scroll wheel	Step the value up or down.
Double-click	Reset to the default value.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number.

In MAP mode, with a source armed, each knob that source moves draws a **route ring** in its colour outside its own arc, spanning how far the source moves it. Outside MAP the knobs show no rings; the route list shows every route. In MAP mode, knobs behave differently: see *Chapter 13*.

Colours

Part A is drawn in ● orange and part B in ● blue, in the sky and on the panels. Every modulation source has its own colour, the same on its chip, its route rings and in the route list, and MAP and the route list are ● pink.

Panel size

The panel is 1648 × 940 at 100%. Right-click any empty part of the sky and choose *UI Size* to scale it to 75%, 100%, 125%, 150% or 200%; at 75% it fits a 1440 × 900 laptop screen. The size you pick is remembered the next time the editor opens.

CHAPTER 05

How JSynth Works

A note becomes a voice and a creature at once. The voice makes the sound; the creature flies, and how it flies can shape the sound.

One voice

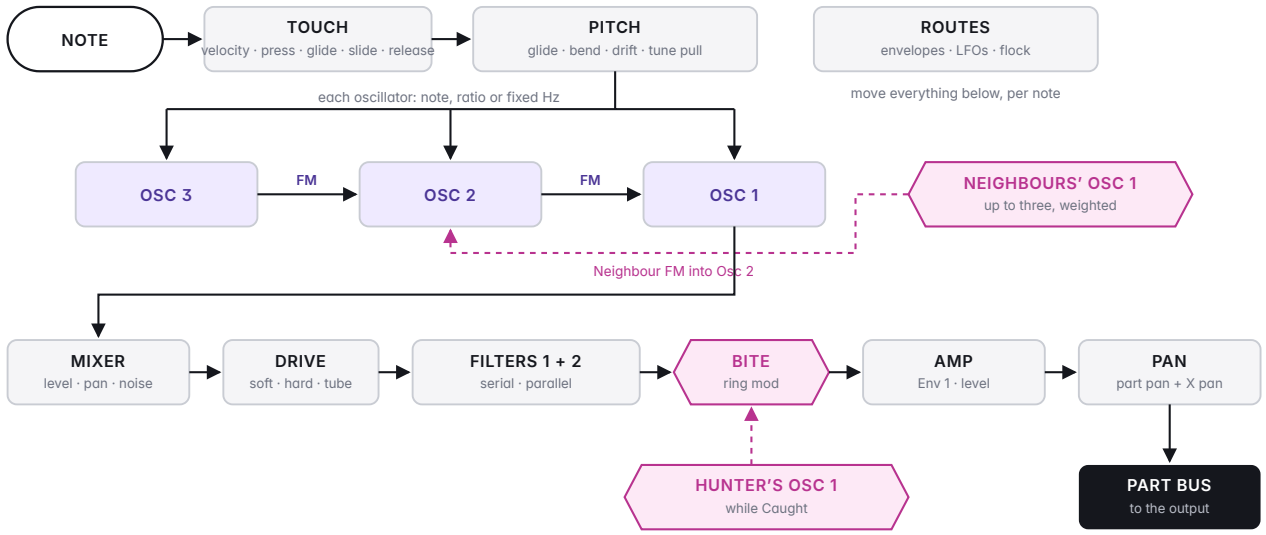


Fig. 5.1 One voice, at the oversampled rate. The pink paths are the couplings to other creatures: a neighbour's Osc 1 into Osc 2, and a hunter's Osc 1 ringing the voice.

Each voice runs three **oscillators**, in the order 3, 2, 1, so Osc 3 can modulate Osc 2, and both can modulate Osc 1, within the same sample. The **FM grid** sets how much each one modulates each other one. The **mixer** sets each oscillator's level and pan and adds noise. Then comes **drive**, then two **filters**, then the **amp**, whose shape is Env 1, and finally the voice's place in the stereo field. Every voice of a part sums to that part's bus, and the two parts sum to the output.

The control rate

Envelopes 2 and 3, the LFOs, the touch, the routes and the flock all work at a control rate of one tick every 32 samples (1,500 a second at 48 kHz). Every value the audio reads ramps smoothly across each tick, so nothing zippers, and knobs are smoothed over 10 ms. The amp envelope runs every sample. Because the ticks are counted across the host's blocks, JSynth sounds the same at any buffer size.

The creature

As a voice starts, its creature hatches at the note's lure. The flock steps 500 times a second. Each step, every creature feels its neighbours (cohesion, alignment, separation), its lure, its own wander, the wind and its wingbeat, and moves. Then the flock works out each creature's four sources and who can hear whom. Each voice reads its creature's sources smoothly between steps.

Routes

A **route** connects a source to a destination with an amount. Each part has 64 of them. Each tick, every voice works out its routes from its own sources, so the same route can push one note's filter open and leave another's closed: an envelope, a finger's pressure or a creature's speed is always that note's own. A route moves a parameter within its own range, so +100% from a source at full takes any knob from its lowest to its highest.

Quality

The voices run at 2× the host's rate by default, which keeps FM and hard waves clean, and are brought back down by a steep, latency-free filter. The **Quality** parameter (a host setting) chooses 1×, 2× or 4×. 1× saves CPU; 4× is for bright FM at low sample rates.

> **WHERE TO READ MORE**
> `plugins/jsynth/ARCHITECTURE.md` in the source repository walks through every stage with the numbers the code uses.

CHAPTER 06

The Sky

Both species fly here, all the time. The sky is the synth's own flock, drawn as it is.



Fig. 6.1 Part A's creatures and part B's in a layered patch, with their ink, in the Sumi look. Ripples are lures; a pink ring marks a creature a hunter has caught; dotted lines join neighbours that hear each other.

What you see

Creatures In Sumi a brush stroke, thick at the head and tapering behind, its width breathing with its Wingbeat LFO; in Aurora a four-pointed star trailing a thin line, its points twinkling with the wingbeat. It is brighter the harder you press, and fades with its sound after you let go. The part you are editing flies in front; the other is dimmed.

Ink Every creature sheds ink behind it, which fades by the part's **TAIL** and spreads by its **SPREAD**, taking on a deeper colour as it ages. In Sumi it marbles into fine filaments; in Aurora each wisp hangs a curtain of light above it. Creatures flying through ink stir it, and so does the mouse.

Lures Where each held note pulls its creature: ripples in Sumi, wider the harder you press, and a small diamond over a faint plumb line in Aurora. **RINGS** under the sky shows or hides them.

Links	From a creature to the neighbour it hears most, brighter the more **FLOCK FM** it gets: dotted in Sumi, an arc with a pulse running along it in Aurora. A hunter has a line to what it has caught.
Readout	Hover a creature to read which voice it is and its Speed, Turn and Nearest.
Caught	A pink ring around a creature a hunter holds.

Looks and palettes

The switch over the sky's top left picks how it is drawn. **SUMI** is ink in dark water, lit from above; **AURORA** is a night sky over a glowing horizon. The menu beside it picks a palette: **EMBER**, **BOREALIS**, **GLACIER**, **SUNSET**, **MOONLIGHT**, **NEON** or **CLASSIC**, each with a set of inks for each part. Each look keeps its own palette (Sumi starts in Ember, Aurora in Borealis), and the look and both palettes are saved with your project. They are only drawn and never change the sound.

Playing the sky

Click anywhere in the sky to play a note: left to right runs from C2 to C6, and how high you click sets Slide. Drag while holding to bend the note and change Slide; let go to release it. The notes play on whichever parts the play mode sends them to, just like notes from your keyboard.

The flock sources

The row of chips under the sky holds each creature's four sources, and **RANDOM**, a new value for each note. Each chip's light shows the newest creature of the part you are editing. They are always each note's own when used in a route.

SOURCE	WHAT IT MEASURES
SPEED	How fast the creature flies, against the fastest it could at full Press (0 to 100%).
NEAREST	How near its nearest neighbour is: 100% is touching, 0 is at the edge of **RANGE** or beyond.
HEADING	How much it flies upward: 100% straight up, 50% level, 0 straight down. Smooth all the way round, so it never jumps.
TURN	Which way and how hard it is turning: up to +100% for a hard right turn on screen, down to −100% for a hard left, 0 flying straight. Smoothed over about 170 ms, so the wander's jitter and the wingbeat's bob barely reach it.

Two views

The switch at the top right picks how much of the panel you see. **SYNTH**, as this manual shows it, puts the sound down the left beside the sky. **SKY** folds everything away but the switch and the look so the sky fills the panel, which is lovely on a second screen while you play. The view stays as you left it when the editor is closed and opened again.

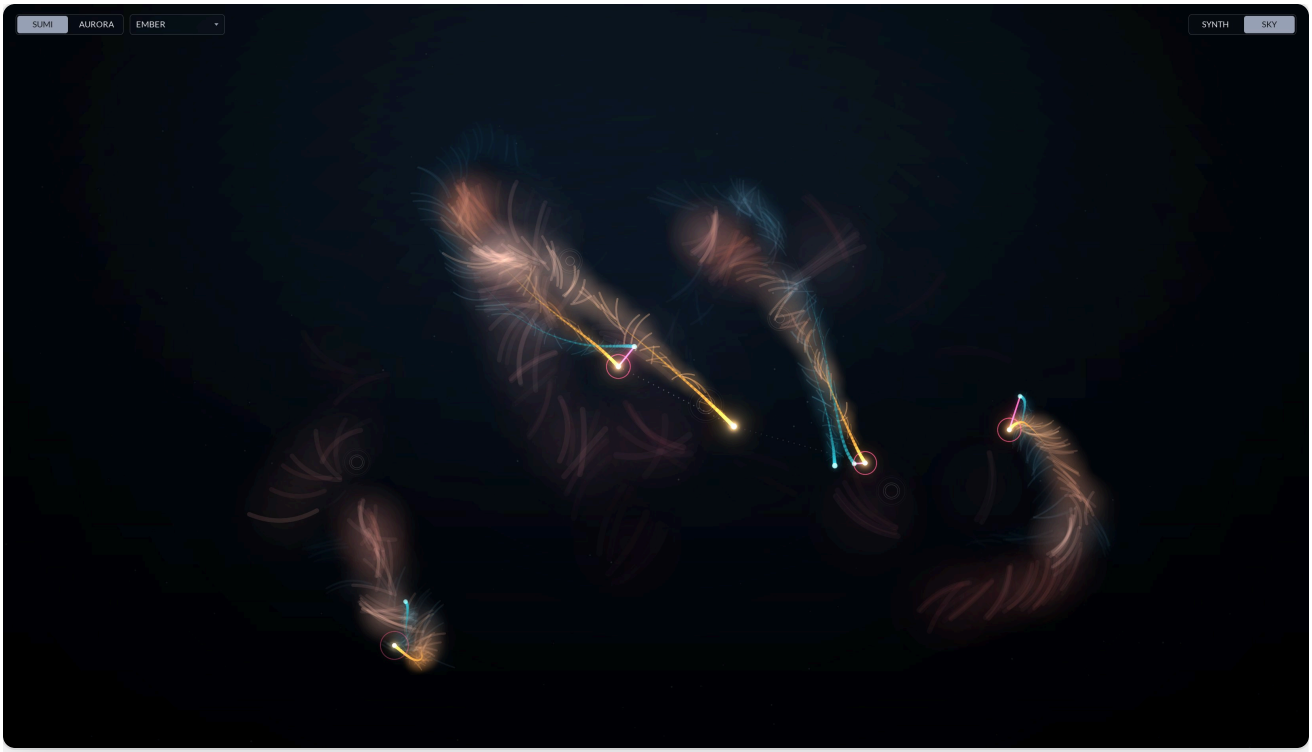


Fig. 6.2 SKY: the flocks alone.

CHAPTER 07

Parts and Playing

Two species, one keyboard. The header decides which notes go where; the VOICE tab under the sky sets how each part plays them.



Fig. 7.1 The header: logo, preset bar, PART A / PART B, play mode, MAP, the touch chips and the view switch.

Two parts

Parts A and B are complete, separate patches with the same controls. **PART A** and **PART B** in the header choose which one the panels show and edit; the sky always shows both. A new instance starts with A shoaling and B hunting it an octave down, so layering them gives you a predator from the start.

Play modes

MODE	WHICH PART PLAYS A NOTE
SINGLE	Part A only. The default.
LAYER	Both, every note.
SPLIT	B below the **Split Key** (C4 by default), A from it up.
MPE ZONES	MPE member channels 2 to 8 play A and 9 to 16 play B. Channel 1 plays both.
VELOCITY	B for notes struck at or above the **Velocity Split** (80), A for softer ones.
EXPRESSION	Both, crossfaded per note by Slide or Press (the **Crossfade Source**): at 0 you hear A, at full you hear B.

Split Key, Velocity Split and Crossfade Source, and each part's Key Low and Key High range, are host parameters. A note outside a part's key range never plays that part.

VOICE



The VOICE tab draws what its knobs set: the stereo field (the part's pan, its X Pan spread, the output's width and where its creatures sit), the voices in use, and at the right the **meters**. These show the levels of A, B and the whole output as they play, from −60 to +6 dB, glowing brighter the louder they are and red above 0 dB. A white tick on each marks where its LEVEL or MASTER is set. The part you are editing is lit; the other is dimmed.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLY / MONO / LEGATO	`Voice mode`	`Poly`	Mono plays one note at a time and restarts it; Legato glides to a new note without restarting while one is held. Both go back to the last note still held when you let go.
Voices (– 8 +)	`1 – 16`	`8`	Polyphony: how many notes the part sounds at most. The oldest released note is taken first, then the oldest held, with a 3 ms fade.
PAN	`±100%`	`0`	The part's pan. ROUTE
X PAN	`0 – 100%`	`0%`	How far each creature's place in the sky, left to right, pans its voice. At 100% a creature at the left edge plays hard left. ROUTE
WIDTH	`0 – 200%`	`100%`	The whole output's stereo width: 0% is mono, past 100% wider. Shared by both parts.
GLIDE	`0 ms – 20 s`	`0`	Portamento time. ROUTE
BEND	`1 – 96 st`	`48 st`	The MPE bend range: how far a finger slides on a member channel.
TRANSP	`±48 st`	`0 (B: -12)`	Transpose, in semitones.
VELOCITY	`0 – 100%`	`100%`	How much velocity sets the level. At 0 every note plays at full level.
LEVEL	`-60 – +6 dB`	`0 dB`	The part's level. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MASTER	`-60 – +6 dB`	`-6 dB`	The whole output's level. Shared by both parts.

Host parameters that complete the part: **Fine** (±100 cents), **Glide Mode** (Always, or Legato: glide only from a note still held), **Key Low** and **Key High**, and **Output** (Main or Own).

MPE and MIDI

JSynth takes CLAP notes, plain MIDI and MIDI MPE.

- **Channel 1** is the MPE manager channel and a plain keyboard's channel. Its pitch bend moves every note by the **MGR BEND** range (2 semitones by default, on the TOUCH tab). CC 1 is the Mod Wheel source and CC 64 is sustain. If it sends channel pressure or CC 74, those reach every note too.
- **Channels 2 to 16** are member channels, each carrying one note's own bend (by the part's **BEND** range), pressure and CC 74.
- **CLAP notes** use note expressions instead: tuning, pressure and brightness.

Hosts that support CLAP per-note modulation can also modulate any parameter tagged ROUTE on a single note.

CHAPTER 08

Oscillators and FM

Three oscillators per voice, from soft sines to band-limited saws, and a grid that lets each one modulate the others.



Fig. 8.1 The OSCILLATORS panel: a row for each oscillator, with its wave drawn as its Wave and Shape make it.

An oscillator

CONTROL	RANGE	DEFAULT	WHAT IT DOES
NOTE / RATIO / HZ	`Pitch mode`	`Note (Osc 2, 3: Ratio)`	**Note** follows the keyboard with an offset; **Ratio** runs at a multiple of the note, the classic FM way; **Hz** holds a fixed frequency whatever you play.
Wave	`15 waves`	`Saw (Osc 2, 3: Sine)`	See the table below.
RAND / SYNC	`Retrigger`	`Sync`	**SYNC** starts each note's wave at its **Phase**; **RAND** at a random place, so no two notes start alike.
PITCH	`±48 st`	`0`	Note mode: the offset in semitones. ROUTE
RATIO	`×0.25 – ×16`	`×1 (Osc 3: ×3)`	Ratio mode (the knob reads RATIO): the multiple of the note. Routes move it by musical intervals. ROUTE
FREQ	`1 Hz – 10 kHz`	`440 Hz`	Hz mode (the knob reads FREQ): the fixed frequency. ROUTE
FINE	`±100 ct`	`0`	Fine tune, in every mode. ROUTE
SHAPE	`0 – 100%`	`50%`	Pulse width for Square and Pulse; for the other waves (except Saw), a phase distortion that squeezes or stretches the first half of the cycle. 50% is the plain wave. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
LEVEL	`0 – 100%`	`100% (Osc 2, 3: 0%)`	How loud it is in the mix. An oscillator at 0 can still modulate others. ROUTE
PAN	`±100%`	`0`	Its place in the voice's stereo image. ROUTE
FILTER	`1+2, F1, F2`	`1+2`	Which filter it goes into. See *Routing* in Chapter 09.

Host parameters per oscillator: **Sync** (Osc 2 and 3 restart with Osc 1), **Phase** ROUTE, **Bend** (how much the pitch bend moves it, 100% by default) and **Key Track** (Off pitches it from middle C whatever you play).

The waves

WAVE	CHARACTER
Sine	Pure. The classic FM operator.
Triangle	Soft, odd harmonics.
Saw	Bright and full. Band-limited, so it stays clean high up and under FM.
Square, Pulse	Hollow to nasal. Shape sets the width: 10 to 90% for Square, 5 to 50% for Pulse.
Half Sine … Alt Camel	The Yamaha TX81Z's seven other waves: Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel and Alt Camel. Sines with parts of the cycle cut, folded or doubled, which give FM a whole family of brighter, buzzier tones.
White, Pink	Noise, as an oscillator: it can be filtered, panned and even used as an FM source.
Random	A different one of Sine to Alt Camel for each note, so a run of notes changes colour as it goes.

OSCILLATORS

FM



The **FM** tab draws what comes out of Osc 1, the oscillator you hear on its own, for a C4, with a chip for each FM cell that is turned up. The grid reads **row into column**: the knob in row 2, column 1 (**2 > 1**) is how much Osc 2 modulates Osc 1. The diagonal, **FB 1** to **FB 3**, is each oscillator's feedback on itself. Each knob is squared inside, so the first half of its travel is gentle.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
FB, 1 > 2 … 3 > 2	`0 – 100%`	`0%`	The nine cells. ROUTE
PHASE / EXP	`FM type`	`Phase`	**Phase** is DX-style phase modulation, bright and stable in pitch. **Exp** moves the pitch itself, up to four octaves either way: growly, and it bends.
FLOCK FM	`0 – 100%`	`0%`	Neighbours' Osc 1 into this creature's Osc 2. ROUTE
TUNE PULL	`0 – 100%`	`0%`	Creatures flying the same way drift toward pure intervals. ROUTE
BITE	`0 – 100%`	`0%`	Ring mod while this part's hunters hold their prey. ROUTE
NOISE	`0 – 100%`	`0%`	The mixer's noise level. ROUTE
COLOR	`±100%`	`0`	Pink (−100%) to white (+100%). ROUTE
DRIFT	`0 – 100%`	`0%`	Each creature's own slow wander in pitch, up to ±15 cents. ROUTE

Order and feedback

Within each sample the oscillators run 3, then 2, then 1. So the downward cells (3 > 2, 3 > 1 and 2 > 1) hear this sample's modulator, and the upward ones (1 > 2, 1 > 3, 2 > 3) hear the last sample's. In practice this means you can build any of the classic stacks: 3 > 2 > 1 as a three-operator stack heard on Osc 1, or 2 > 1 and 3 > 1 as two modulators on one carrier. Feedback averages its last two samples, as the DX7 did, so it brightens toward a saw and then a rasp rather than straight into noise.

BETWEEN CREATURES (Flock FM, Tune Pull and Bite) is explained in *Chapter 11*.

CHAPTER 09

Filters and Drive

Two filters per voice, from a clean 48 dB low pass to a ringing comb, after a drive stage.

LEFT

FILTER



The graph draws both filters' response together, Filter 1 in orange and Filter 2 in purple. Drag a numbered handle across to set its cutoff and up or down for its resonance; the wheel sets its drive and a double-click resets it. In MAP mode, with a source armed, the drag sets that source's route to the cutoff instead.

While a note plays, the curves follow what the newest note's filters are really doing: its envelopes, LFOs, routes and key tracking all move them, and a faint line stays where the knobs alone would put them. The handles stay on the knobs' settings.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SERIAL / PARALLEL	`Routing`	`Serial`	See below.
NOISE INTO	`1+2, F1, F2`	`1+2`	Which filter the noise goes into.
Type	`8 types`	`Low Pass (F2: Ladder)`	See the table below.
Slope	`12 – 48 dB`	`24 dB`	Low, high and band pass only.
CUTOFF	`20 Hz – 20 kHz`	`1.26 kHz (F2: 20 kHz)`	For a comb, its tuning. ROUTE
RES	`0 – 100%`	`20% (F2: 0%)`	Resonance; for a peak, its boost; for a comb, its feedback. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
DRIVE	`0 – 100%`	`0%`	Saturates the filter's own input. ROUTE
KEY	`0 – 100%`	`0%`	Key tracking: at 100% the cutoff moves an octave per octave, around middle C. ROUTE
Drive type	`Soft, Hard, Tube`	`Soft`	The drive before both filters.
PRE DRIVE	`0 – 100%`	`0%`	Up to 24 dB into the shape, with the level brought back down. ROUTE
BALANCE	`±100%`	`0`	In Parallel, turns Filter 1 or Filter 2 down. ROUTE

Filter types

TYPE	SOUND
Low Pass	The workhorse: a state variable filter, cascaded for 12, 24, 36 or 48 dB per octave, with a resonance peak up to a whistle.
High Pass	The same, from the other side: thins and brightens.
Band Pass	Only a band around the cutoff. Narrower and more vocal with resonance.
Notch	Cuts a band out. Resonance narrows it. Slope is ignored.
Peak	Boosts a band. Resonance is the boost, up to five times.
Ladder	A four-pole transistor ladder: warm, thick, and it squelches and self-oscillates near full resonance.
Comb +	A short tuned delay with positive feedback: rings at every harmonic of the cutoff, like a plucked string or a tube.
Comb –	Negative feedback: rings at the odd harmonics, hollower, an octave lower.

Routing

Serial Filter 1 into Filter 2. Filter 2 starts as an open ladder, so it changes nothing until you turn it down, and costs nothing while it stays open.

Parallel Each filter hears what goes into it, and **BALANCE** turns one side down.

Each oscillator's **FILTER**, and the noise's, picks where it goes: **F1**, **F2** or both (**1+2**, the default). In Serial, F1 and 1+2 go through both filters and F2 skips Filter 1. In Parallel, a source in one filter plays at full level with BALANCE at centre, and turning BALANCE the other way turns it down. A source in both crossfades: half into each at centre, all into one at either end.

> **TWO SOUNDS IN ONE**
> Set Osc 1 to **F1** and Osc 2 and 3 to **F2**, in Parallel, to filter a body and a bright layer apart: a low pass on one, a comb or high pass on the other.

Drive

The drive sits before the filters. **Soft** is a tanh curve, round and warm. **Hard** clips flat, buzzier. **Tube** is asymmetric, so it adds even harmonics. Its gain is compensated, so turning it up thickens and compresses the sound more than it makes it louder.

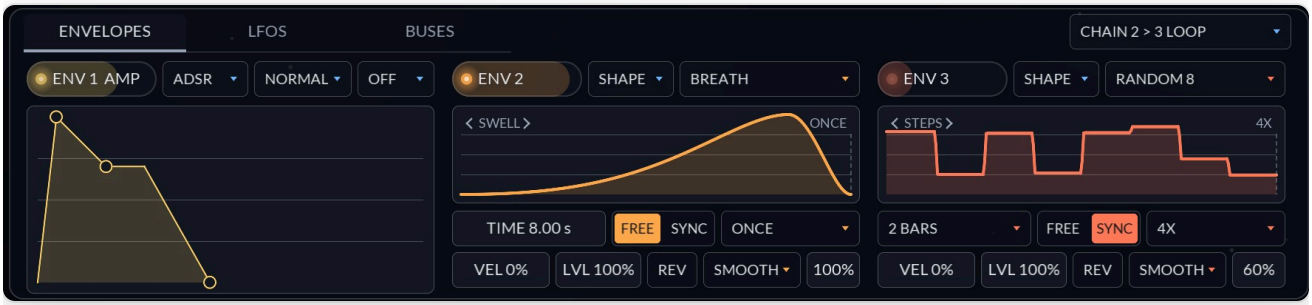
CHAPTER 10

Envelopes, LFOs and Buses

Three envelopes and four LFOs per voice, and four mod buses per part. Two of the LFOs move the creatures as well as the sound.

BOTTOM LEFT

ENVELOPES



Each envelope is an **ADSR** or a **SHAPE**, picked in the box beside its chip. As an ADSR it draws its stages: drag the points to set **attack**, **decay**, **sustain** and **release**. A point moves with the mouse from where you grab it; hold **Shift** for fine moves. A dot rides the curve for every sounding note. The chip above each one is its source for MAP and lights as it moves.

ENV 1 AMP is the amp: it shapes every note's level, and it always releases when you let go. **ENV 2** and **ENV 3** are free to route; a new part starts with Env 2 opening Filter 1.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Attack	`0 ms – 20 s`	`5 ms`	ROUTE
Decay	`0 ms – 20 s`	`400 ms (2, 3: 300 ms)`	ROUTE
Sustain	`0 – 100%`	`70% (2, 3: 30%)`	ROUTE
Release	`0 ms – 20 s`	`300 ms`	ROUTE
Time Range	`Short, Normal, Long`	`Normal`	How much time the drawing spreads across its width: up to 1 s, 5 s or 20 s for each of attack, decay and release. Pick **Short** for plucks and drums, **Long** for pads and swells. It only changes the view: switching it never changes a time, and a time past the end waits there until you drag it back in.
Loop	`Off, Held, Free`	`Off`	**Held** repeats attack and decay while the note is held. **Free** keeps repeating after you let go too, until the note ends.

VELOCITY on the VOICE tab under the sky (0 to 100%, 100% by default) is how much velocity sets ENV 1's level. At 0 every note plays at full level.

Host parameters per envelope: **Delay** and **Hold** (0 ms to 20 s) ROUTE, an **Attack**, **Decay** and **Release Curve** each (−1 to +1), and **Velocity**, how much velocity scales the whole envelope.

Shape mode

After XAOC's Zadar module, a **SHAPE** envelope plays one drawn contour instead of its stages: pick it from **60 shapes** in ten banks, *Rise*, *Fall*, *Swell*, *Pulse*, *Waves*, *Steps*, *Drift*, *Strike*, *ADSR* and *Acoustic*, and it is stretched over its time. Click the shape box for a menu by bank, scroll over it, or use the arrows either side of the bank's name on the drawing to step through them (the name itself opens the menu). Long, slow shapes on a filter or a drawbar keep a held chord moving for minutes; a pulse or a step shape as ENV 1 turns each held note into a pattern of its own.

The drawing shows the shape as it plays, with its bank and how often it repeats in its corners and a dot for every sounding note. **Drag it sideways to warp it** in time and **up or down to bend it** in level (Shift for fine moves; double-click puts both back). In MAP mode, with a source armed, the same drag sets how far that source warps it (sideways) or bends it (up or down), and the shape as the source at its fullest would take it shows over it in the source's colour; right-click removes both routes.

Sustain point. The dashed line on the drawing is where the shape stops and holds while you hold the note, as on Zadar; let go and it plays on from there. Drag it in from the right edge (all the way back to the edge turns it off) and double-click it to turn it off. On ENV 1 the rest of the shape is the release: it plays out first, then the Release fade follows. The ADSR and Acoustic banks draw the classic stages, so a sustain point on their flat part makes an ADSR you can see.

Envelopes moving envelopes. Env 1, 2 and 3 are sources, and each envelope's Time, Warp, Bend, Level, Sustain Point and Chain Phase (and its ADSR stage times) are destinations, so one envelope can stretch, lean or bend another while it plays, or itself. Time is read as a rate, so a change speeds the shape up or slows it from where it is, never jumps it. Chain Phase is read when the chain hands over. The **Interplay** presets are built this way.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Time	`1 ms – 30 min`	`2 s`	How long one pass takes, with **FREE** lit. ROUTE
Length	`1/16 – 64 bars`	`1 bar`	With **SYNC** lit, one pass in the host's tempo instead.
Warp	`-100 – +100%`	`0%`	Drag sideways: + pushes the shape later, – pulls it sooner. ROUTE
Bend	`-100 – +100%`	`0%`	Drag up or down: + lifts the shape, – sinks it. ROUTE
Repeat	`Once, 2x – 16x, Held, Forever`	`Once`	How many passes it plays; after the last it holds where it ended. **Held** repeats while the note is held and finishes its pass after; **Forever** until the note ends.
LVL	`0 – 100%`	`100%`	The shape's height, as Zadar's level: it scales the whole contour down. ROUTE
Sustain	`0 – 100%, Off`	`Off`	The dashed line: where the shape holds while the note is held. ROUTE
REV	`Off, On`	`Off`	Plays the shape backwards. The sustain point stays where it is from the start.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Retrigger	`Smooth, Restart, Ignore`	`Smooth`	What a new note does while it plays: **Smooth** picks the shape up where it passes the level it is at now, so it never jumps; **Restart** starts it from the top, gliding there in 5 ms so it never clicks; **Ignore** lets it play on.
REL / VEL			ENV 1's **Release**: when you let go it fades over this while the shape keeps moving under the fade (with a sustain point, after the rest of the shape has played). ENV 2 and ENV 3 show their **Velocity**, and play on after you let go.

Chains. The box at the top right of the panel chains the envelopes in Shape mode, so one starts the next: **1 > 2**, **2 > 3**, **1 > 2 > 3**, or the same as a **Loop** back to the first. The first starts with the note; each next one waits until the one before has played its **phase** (the box beside its Retrigger) of its whole run: 0% starts it with the one before, 100% when that one ends. In a loop the first starts again from the last the same way, so two arches that start each other halfway cross-fade forever. In a chain, Held and Forever play once, so the chain moves on. An ADSR envelope in a chain just starts with the note.

UNDER THE SKY

MOTION



The MOTION tab holds the two LFOs that move the flock. They only move it: they aren't sources for routes. For a sound that moves with them, use LFO 1 or 2 at the same rate.

WINGBEAT One per creature (5 Hz sine by default). It bobs each creature across its path, by **PUSH**, and beats its wings on screen.

WIND One gust for the whole sky (a slow 0.2 Hz Smooth, Global by default). It pushes every creature of the part the same way, by **PUSH**, from a direction that turns slowly on its own.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Shape	`Sine, Triangle, Saw, Square, S&H, Smooth`	`Sine (Wind: Smooth)`	S&H holds a random value per cycle; Smooth glides between random values.
Trigger	`Retrigger, Free, Global`	`Retrigger (Wind: Global)`	See below.
FREE / SYNC	`Rate or tempo`	`Free`	Sync follows the host's tempo at **DIVISION**; Free and Global also lock to its bar.
RATE	`0.02 – 50 Hz`	`5 Hz (Wind: 0.2 Hz)`	ROUTE
PUSH	`0 – 100%`	`20% (Wind: 30%)`	How hard it moves the creatures. ROUTE
PHASE	`0 – 360°`	`0°`	Where a retriggered cycle starts, or how far a synced one sits off the beat. ROUTE
DELAY	`0 ms – 20 s`	`0`	Fades the LFO in after each note starts. ROUTE
DIVISION	`1/32 – 4/1`	`1/4`	With triplets and dotted values.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLARITY	`Bipolar, Unipolar`	`Bipolar`	Bipolar swings both ways around the knob; unipolar only up from it.

ENVELOPES

LFOS



LFO 1 (4 Hz sine) and **LFO 2** (0.5 Hz triangle) are plain LFOs that only modulate. The **LFOS** tab draws each one, with a dot running along it at its rate. They have the same controls as the MOTION LFOs, without Push.

TRIGGER

Retrigger	Each note's LFO starts at Phase. Notes are in step with how you play them.
Free	Each note's LFO picks up where the part's running LFO is, so notes are in step with each other.
Global	One LFO for the whole part. Every note hears the same value; a route can't change its rate per note.

ON THE GRID

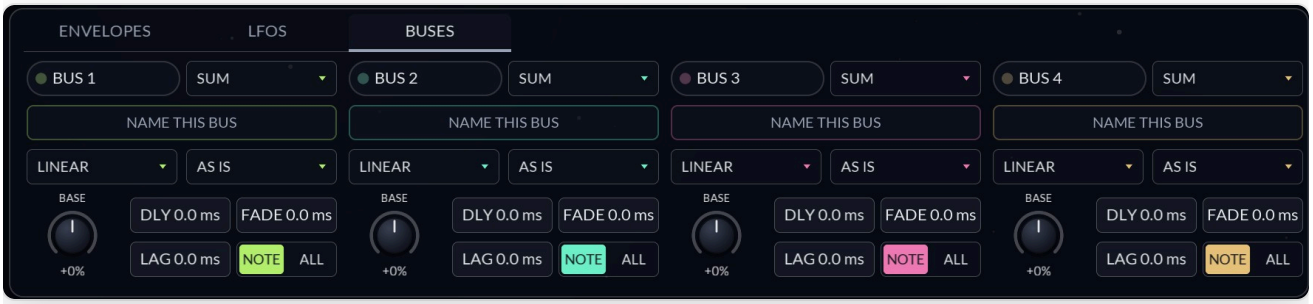
With **SYNC** on, a **Free** or **Global** LFO is locked to your DAW's bar, not just its tempo: a 1/16 gate lands on the sixteenths wherever you start playback, and a looped bar plays the same S&H values every time round. **PHASE** shifts it against the beat. **Retrigger** starts it with each note instead, which suits notes played off the grid.

VIBRATO

Route LFO 1 to the part's **Fine** at a few percent for vibrato, and set its **DELAY** so it arrives once the note has settled. Map Press onto that route's amount and the vibrato follows your finger.

ENVELOPES

BUSES



Each part has four **mod buses**. A bus takes any number of sources, mixes them into one value, shapes it, and hands it on as a source of its own: **BUS 1** to **BUS 4**, with a chip on this tab. Build a mix once (say, Press times the mod wheel, faded in after two seconds) and route it to as many places as you like.

To feed a bus, touch a source in MAP and drag the bus's **BASE** knob, or pick the bus as a route's destination in the route list. To use it, touch its chip and drag any knob. A bus can feed another bus.

HOW IT MIXES AND SHAPES

MODE	**SUM** adds the inputs, each at its route's amount. **MULTIPLY** multiplies them, so one gates or scales another. **MIN** keeps the lowest and **MAX** the highest.
BASE	Added to the mix, −100% to +100%. With no inputs, a bus is just this value.
CURVE	**LINEAR** as it is; **EXPONENTIAL** slow near zero; **LOG** quick near zero; **S-CURVE** eased at both ends; **ABSOLUTE** folds the negative half up.
POLARITY	**AS IS**; **UNIPOLAR** turns −100..+100% into 0..100%; **BIPOLAR** turns 0..100% into −100..+100%; **INVERT** flips it.
DLY, FADE	The bus stays at zero for DLY after each note starts, then rises to full over FADE. This is how you bring in any modulation late in a held note.
LAG	Smooths the bus, so a step glides and a fast source slows down.
NOTE / ALL	**NOTE** works the bus out for each note. **ALL** makes it one value for the part, from its newest note, and times DLY and FADE from the first key down, so a note played into a held chord doesn't start the fade again.

Click the box under a bus's chip to name it. The name shows on its chip, in the route list and in MAP's hint, and is saved with the song and in your presets. An empty name goes back to "Bus 1".

> **LATE VIBRATO**
> Route LFO 1 onto Bus 1 at 100%, set the bus's DLY to 2 s and FADE to 1 s, and route Bus 1 to Fine at a few percent. Any held note now grows vibrato after two seconds. Route Press onto Bus 1 as well with MODE on MULTIPLY, and your finger sets how deep it gets.

CHAPTER 11

The Flock

What boids are, how each species flies, and how the way a creature flies becomes how it sounds.

What boids are

JSynth's creatures are **boids**, the flocking model Craig Reynolds published in 1986. No creature knows where the flock is going, and nothing steers the flock as a whole. Each creature looks only at its *neighbours*, the creatures of its own species within **RANGE**, and follows three rules:



Fig. 11.1 The three rules, for the orange creature. The dashed circle is **RANGE**, its neighbourhood; the inner pink one, 40% of Range, is too close. Each rule is a pull (pink) whose strength is its knob.

Cohesion Steer toward the middle of your neighbours. This is what holds a flock together.

Alignment Match your neighbours' heading and speed. This is what makes a flock turn as one.

Separation Steer away from any neighbour that gets too close. This is what keeps a flock from collapsing to a point.

Run those rules on every creature, 500 times a second, and shoals, swirls and murmurations come out of nothing. Nearer neighbours count more: a neighbour's pull fades smoothly to nothing at the edge of Range, and it counts for less while its own sound is fading, so a creature drifting in or out of the neighbourhood never jolts the others. JSynth adds forces of its own on top of the three rules: your note's lure, each creature's wander, the wind and the wingbeat, the walls, and a behaviour that decides how each species treats the other.

A creature's life

Each part is a **species**, and every sounding voice is one creature of it. A creature flocks only with its own kind; the other species matters only when one of them hunts or flees.

Hatching When a note starts, its creature hatches at the note's lure, pointing a random way. The sky is empty until you play.

Held While the note is held, its creature flocks with the others and is pulled toward its lure. It glows brighter the harder you press, and Press also raises its top speed.

Released When you let go, the lure lets go too. The creature flies on with the flock as its sound fades, dimming as its amp envelope releases.

Gone When the voice ends, the creature fades out and its slot is free for the next note. A stolen voice's creature goes the same way.

One step of the flock

500 times a second, each creature adds up these pulls, turns toward the sum, and moves. Its speed always stays between a fifth of its top speed and the top speed itself.

FORCE	WHAT IT DOES	SET BY
Cohesion	Toward the middle of its neighbours (away from it on Scatter).	COHESION, RANGE
Alignment	Toward its neighbours' average heading and speed.	ALIGN, RANGE
Separation	Away from any neighbour inside 40% of Range, harder the closer it is.	SEPARATE, RANGE
Lure	While the note is held, toward a point that roams around the lure on a leash: loose near it, firm when the creature strays.	LURE, WANDER, LURE X/Y
Wander	A heading of its own that drifts, so it sometimes sweeps a long curve and sometimes jinks.	WANDER
Hunt, flee	A hunter chases the nearest creature of the other species; prey shies from hunters; a fleeing species runs from the other.	Behaviour
Wind	One gust pushing every creature of the species the same way, from a direction that turns slowly.	WIND (MOTION tab)
Wingbeat	Bobs each creature side to side across its path.	WINGBEAT (MOTION tab)
Walls	Turns creatures back at the sky's edges, or carries them round.	WALLS

UNDER THE SKY

FLOCK



CONTROL	RANGE	DEFAULT	WHAT IT DOES
Behaviour	`Shoal, Scatter, Hunt, Flee`	`Shoal (B: Hunt)`	How this species treats its own kind and the other. See Behaviours below.
COHESION	`0 – 100%`	`50%`	Rule 1: how hard each creature steers toward the middle of its neighbours. ROUTE
ALIGN	`0 – 100%`	`50%`	Rule 2: how hard each creature matches its neighbours' heading and speed. ROUTE
SEPARATE	`0 – 100%`	`40%`	Rule 3: how hard each creature steers away from a neighbour inside 40% of Range. ROUTE
SPEED	`0 – 100%`	`40% (B: 55%)`	Top speed. Press raises it further, up to 80% more. ROUTE
WANDER	`0 – 100%`	`20%`	Each creature's restlessness, and how far it roams around its lure. ROUTE
LURE	`0 – 100%`	`60%`	How hard a held note pulls its creature. Low lets the flock take over. ROUTE
RANGE	`0 – 100%`	`35%`	How near counts as a neighbour: 8% to 40% of the sky's height at the default Size. ROUTE
TAIL	`0 – 100%`	`50%`	How long the ink lasts, about half a second to five seconds. ROUTE
SPREAD	`0 – 100%`	`50%`	How far the ink blooms as it ages. ROUTE
SIZE	`0 – 100%`	`50%`	How big the sky is. A bigger sky makes creatures, their Range and their speed relatively smaller, so flocks spread out. Shared.
WALLS	`Soft, Wrap, Open`	`Soft`	Soft turns creatures back near the edge; Wrap carries them round to the other side; Open lets them fly well past it first. Shared.
LURE X, LURE Y	`Pitch, Velocity, Press, Slide, Release, Random, Centre`	`Pitch, Slide`	What places a note's lure along each axis.

Behaviours

Shoal Flies with its own kind: cohesion, alignment and separation as set. The flock to start from.

Scatter Keeps apart: cohesion pushes creatures away from their neighbours' middle instead of toward it, so a chord spreads across the sky and Nearest stays low.

Hunt Chases the nearest creature of the other species it can see, up to 2.5 times its Range away. A hunter that gets within reach (about 6% of the sky's height) catches it, which turns on Bite. Every creature shies away from a hunter of the other species a little, so prey scatters as a hunter comes in.

Flee Runs hard from any creature of the other species within its Range, whatever that species is doing.

Behaviour only matters between species, so it needs both parts playing: play mode **LAYER**, or any mode that sends notes to both. A new instance starts with A on Shoal and B on Hunt.

The lure

A held note pulls its creature toward a point that roams slowly around the lure, on a leash: loose near the lure, so the flock and the wander shape the path there, and firm when the creature strays. Each creature roams on its own two slow orbits, so no two trace the same path. **LURE X** and **LURE Y** choose what places the lure: by default low notes sit left and high notes right (Pitch, over four octaves from C2, and bend moves it), and more Slide sits higher. Try Lure Y on *Press* to make pressing harder send creatures upward, or Lure X on *Random* to scatter a chord across the sky. A lure follows its note as it plays, so sliding or pressing moves it live.

How the flock reaches the sound

The flock changes nothing in the sound by itself, so a plain patch is plain however it flies. It gets in three ways, and only these:

Sources	Each creature is a modulation source for its own note: **SPEED**, **NEAREST**, **HEADING** and **TURN** (Chapter 6). Touch one under the sky and drag any knob, as with an envelope.
Between creatures	**FLOCK FM**, **TUNE PULL** and **BITE** on the FM tab let creatures hear each other (below).
X Pan	**X PAN** on the VOICE tab pans each voice by where its creature is, left to right.

So the flock's knobs shape the sound through those. This is what each one does to what you hear:

TURN UP	THE CREATURES	WHAT YOU HEAR
COHESION	Bunch together.	Nearest climbs and more creatures are in each other's range, so Flock FM rings harder and Tune Pull has more to pull toward. With X Pan, the stereo image narrows.
ALIGN	Fly as one, turning together.	Heading and Turn move together across a chord, so routes from them move every note alike. Tune Pull engages, since it only pulls creatures heading the same way.
SEPARATE	Keep their distance.	Nearest stays lower and steadier, and Flock FM gentler.
SPEED	Fly faster.	Every source moves faster: neighbours come and go sooner and X Pan sweeps quicker. The Speed source itself is measured against the top speed at full Press, so a creature you don't press stays below about 55%, and pressing fills it.
WANDER	Drift, jink and roam wider.	Turn gets busy and Heading wanders, so routes from them get livelier; the flock holds together less.
LURE	Stay where you play.	Each creature's place follows its note, so X Pan follows Lure X (by default the pitch, low left and high right). Turn it down and the flock carries creatures off.
RANGE	See farther.	The three rules, Nearest, Flock FM and Tune Pull all reach farther, so creatures hear each other from further apart.
SIZE	Spread out in a bigger sky.	Fewer neighbours, lower Nearest, slower sweeps. Shared by both species.
Behaviour	Shoal, scatter, chase or flee.	Shoal gives a high, steady Nearest; Scatter a low one and a clean sound; Hunt and Flee give sudden bursts of Speed and Turn, and Hunt drives Bite.
WALLS	Turn back at the edge, or wrap round.	With Wrap, a creature leaving one side reappears at the other, so X Pan jumps across the stereo field. Soft and Open sweep it back.

TURN UP	THE CREATURES	WHAT YOU HEAR
TAIL, SPREAD	Leave longer, wider ink.	Nothing. Ink is only drawn.

Modulating the flock

Every flock knob but Size can take routes, and a route moves each creature by its own note. A new part already routes Press onto Speed, so the note you lean on flies faster. Try Press onto **SEPARATE** and the note you press breaks formation while the rest hold together; Velocity onto **WANDER** and hard-struck notes fly wild; an envelope onto **LURE** and each creature is held at its place as the note starts, then let loose into the flock. Behaviour, Lure X and Lure Y, Size and Walls can't be routed.

The **MOTION** tab's two LFOs (Chapter 10) move only the flock. The wingbeat shows a little in Turn, and the wind in Speed and Heading, so route those to hear them.

Between creatures

Three knobs on the **FM** tab connect the flock to the sound directly. All start at 0, so the flock changes nothing until you ask it to.

FLOCK FM Each creature listens to up to three neighbours of its own species, weighted by how near they are, and their Osc 1 frequency-modulates its Osc 2. A tight flock rings; a loose one is clean. When a creature gains or loses a neighbour, the change fades in over about 10 ms, so it never clicks. For it to be heard, Osc 2 must be in the sound: give it level, or route it into Osc 1 on the FM grid. It needs at least two notes of the same part.

TUNE PULL For each neighbour heading within 30° of the same way, the creature is pulled toward the nearest just interval from that neighbour's pitch (unison, minor third, major third, fourth, fifth, major sixth, octave), up to ±30 cents at 100%, over about 150 ms. Chords played on a flock that flies together settle into just intonation; break formation and they drift back.

BITE While one of this part's hunters holds a creature of the other species, the hunter's Osc 1 ring-modulates it, by Bite times how caught it is. Set Bite on the hunting part. A catch takes hold in about 20 ms and lets go over half a second, so each bite rings out as the prey escapes.

> **HEAR THE RULES ONE AT A TIME**
> Load *Init*, set OSC 2's LEVEL to half, FLOCK FM to 60% and LURE to 20% so the flock can take over, and hold a four-note chord across two octaves. Turn COHESION to 100% and the others to 0: the creatures pile together and the chord rings. Swap to SEPARATE at 100%: they push apart and it goes clean. Then ALIGN alone: they stream off together, and with TUNE PULL up the chord slides into just intonation as they line up.

Ink

Every creature leaves ink behind it in the sky, more the brighter it is. The ink fades with **TAIL** and spreads with **SPREAD**, and creatures flying through it stir it. How it looks depends on the sky's look and palette (*Chapter 6*). It is only drawn: it doesn't change the sound.

CHAPTER 12

Touch

Five dimensions of every note, each shaped and smoothed, each a source.

UNDER THE SKY

TOUCH



TOUCH	WHERE IT COMES FROM
VELOCITY	How hard the note was struck: its velocity.
PRESS	Pressure: MPE channel pressure or the CLAP pressure expression. It also makes the creature faster and brighter.
GLIDE	The finger's bend, as −100% to +100% of the bend range.
SLIDE	Up and down the key: MPE CC 74 or the CLAP brightness expression.
RELEASE	How fast the note was released: its release velocity, 0 while the note is held. Route it to the amp envelope's release, say, so a quick lift cuts the note short.

Each has a **CURVE** (−1 to +1): positive bows it up, so it responds more early in the travel, and negative bows it down. And a **SMOOTH** time (8 ms by default, 2 ms for Glide so slides track the finger), which irons out a controller's steps.

Under them, the rest of how you play. The whole tab is shared by both parts.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ROUND	`0 – 100%`	`0%`	Glide Rounding: once a finger stops sliding, pull its pitch toward the nearest semitone, by this much. While the finger moves, the pitch follows it exactly.
ROUND TIME	`0 ms – 20 s`	`150 ms`	How long that pull takes.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MGR BEND	`1 – 96 st`	`2 st`	The bend range of the MPE manager channel (channel 1), which bends every note.

The touch chips

The five chips in the header light as you play, showing the newest note of the part you are editing. In MAP mode they are sources like any other: touch **PRESS** and drag a knob to make pressure move it.

> **NO MPE CONTROLLER?**
> Velocity comes from any keyboard. A plain keyboard's channel pressure is Press for every note, and its pitch wheel bends every note by MGR BEND. In the sky, a click sets Slide by how high you click and dragging bends.

CHAPTER 13

Modulation: MAP and Routes

There is no matrix page. Touch a source, drag a knob.

Mapping

01 **Touch a source**

Click any source chip: the touch chips in the header, the envelope, LFO and bus chips on their panels, or the flock's under the sky. That turns MAP on: the button lights pink, the source chips pulse, the one you touched lights, and the sky tells you to drag a knob. (Pressing **MAP** first works too.)

02 **Drag a knob**

Drag any knob up or down to set how much the source moves it, from −100% to +100%. A ring in the source's colour shows how far it reaches. Drag more knobs to send the same source to them too, or touch another source.

03 **Press MAP again**

To finish. Knobs go back to setting their own values, and the rings go away.



Fig. 13.1 MAP with a source armed: the knobs it moves show rings in its colour.

IN MAP MODE, WITH A SOURCE ARMED	ON A KNOB
Drag	Set that source's amount on this knob. A route dragged back to 0 is removed.
Double-click	Flip the route's sign.
Right-click	Remove the route. With no source armed, removes every route to that knob.

The value boxes (an envelope's TIME, REL, VEL and phase) take routes the same way: drag up or down, and a bar along the box's foot shows how far the source moves it. A Shape envelope's drawing takes a sideways drag as a route onto its Warp and an up-or-down one onto its Bend.

Only the armed source's rings show; touch another source to see its routes. Nearly every knob on a part's panels can take routes: the flock's knobs, the FM grid, the envelopes and LFOs, the filters and the oscillators. The exceptions are Transpose, the part's Bend range, Velocity, and the shared ones: Size, Width, Master and the TOUCH tab.

Sources

SOURCE	VALUE
Env 1, 2, 3	0 to 100%, each note's own.
LFO 1, 2	−100% to +100% (or 0 to 100% when Unipolar). Per note unless Global.
Velocity, Press, Slide	0 to 100%, each note's own.
Release	The note's release velocity, 0 to 100%: 0 while it is held.
Glide	−100% to +100%: the note's bend.
Key	The note: −100% four octaves below middle C, 0 at middle C, +100% four octaves above.
Speed, Nearest, Heading	The flock's own, 0 to 100% (Chapter 6).
Turn	The flock's turning, −100% (left) to +100% (right) (Chapter 6).
Random	A new value per note, −100% to +100%.
Mod Wheel	CC 1 on channel 1.
Macro 1 to 4	Four host parameters, for automation or a controller.
Bus 1 to 4	The part's mod buses, −100% to +100%, by their own names (Chapter 10).

Key, Mod Wheel and the macros have no chip on the panel; choose them in the route list. Random's chip is under the sky with the flock's.

The route list



Fig. 13.2 ROUTES: every route of the part in use, with its source, destination and amount.

ROUTES under the sky opens, over the sky, the list of the part's routes in use, sixteen to a page. Each row has the source, the destination and the amount; change any of them here. **+ ADD** starts a new route (Env 2 onto Filter 1 Cutoff, at 0) to edit, and **X** removes one. The arrows turn the pages. Each part has 64 routes; the list shows how many are used.

The list reaches what MAP can't: sources without a chip, and destinations with no knob on the panel, such as an oscillator's Phase, an envelope's Delay and Hold, and every route's own amount.

Routes onto routes

A route's amount is itself a destination. Route Press onto the amount of a route from LFO 1 to Fine, and the vibrato deepens as you press. Routes onto amounts are worked out first, so they always take effect on the same tick.

How far a route moves a knob

A route moves a parameter within its own range: +100% from a source at full moves a knob from its lowest setting to its highest, and +10% moves it a tenth of the way. Routes add together, and the result stops at the knob's ends. Ratio moves by musical intervals, so a small route onto Ratio detunes evenly at every setting.

CHAPTER 14

Factory Presets

One hundred and sixty-six sounds in ten categories: a few plain patches by instrument family first, then the showcases: the flock playing the sound, Press shaped by the mod buses, patches played with pressure, drones and textures on Shape envelopes, patches that move on their own, envelopes that move each other, landscapes on both parts, and variations on MicroMonsta 2 programs.

The preset bar works like JVerb's: the arrows step through the presets, scrolling over the bar does too, and clicking the name opens the menu by category. **SAVE** saves the current sound under a name of your own. The **star** beside the name makes the current preset a favorite, and *Favorites* at the top of the menu lists every starred one, so the sounds you reach for are one click away. A factory preset sets every parameter it doesn't mention to its default, so nothing from the last sound (a route, say) carries over. *Essentials* comes first, a few plain patches to start from; the rest are showcases. Most answer the mod wheel and Press. Every preset is level-matched, so stepping through the bank doesn't jump.

26 PRESETS

Essentials

A few plain patches to start from, by instrument family.

Sub Bass	A sine with a touch of feedback and a quiet octave; the wheel adds feedback.
Analog Bass	Saw and a square an octave down through a ladder; Env 2 plucks it, harder when struck harder.
FM Bass	A DX bass: 1:1 FM with feedback on the modulator, its depth dying away; Velocity bites.
Acid Bass	Saw into a resonant, driven ladder with glide between held notes; Velocity accents, Slide opens it.
Reese Bass	Two saws 22 cents apart over a sub, LFO 2 drifting the detune; drive.
Saw Lead	Two detuned saws, vibrato after 350 ms.
Sync Lead	Osc 2 synced to a silent Osc 1; Env 2, the wheel and Press sweep its pitch.
Pressure Lead	Press opens 2:1 FM and brings in vibrato; Slide adds feedback grit.
Tine Piano	A DX electric piano: a 1:1 pair and a ×14 tine, both harder when struck harder; high notes die sooner.
Concert Grand	Layer: struck strings (two beating saws whose filter the hammer throws open, then closes as the note rings) over a pure FM body; low notes left, high right.
Wurli Drive	The reed piano with Press on the drive: lean in and it snarls; the wheel adds tremolo.
Clav	Narrow pulse through a resonant filter, Velocity on the cutoff.

Tonewheel Organ	Drawbars 16′, 8′ and 5 1/3′, key click, and a rotary speaker the wheel speeds up.
Vibraphone	×4 FM with a tremolo the wheel deepens.
Tubular Bells	A ×3.5 modulator over a long ring, a second carrier an octave up.
Synth Pluck	Saw and detuned pulse through a resonant filter; Slide on the resonance, the wheel lengthens it.
Nylon Guitar	Saw through a key-tracked low pass, Env 2 on the cutoff.
Kalimba	×3 FM blip with an inharmonic ×5.95 partial.
Synth Brass	Two detuned saws, Env 2 on the filter (deeper when struck harder), a pitch scoop, vibrato on the wheel.
String Machine	Three saws with slow detune LFOs; Press and the wheel open the filter.
Solo Cello	Mono saw and pulse with glide; Press brings in vibrato and opens the filter.
Warm Analog	Saw pair through a low pass, slow Env 2.
Glass Pad	×3 and ×0.5, LFO 1 on the FM.
Talk Box	Mono saw and pulse through two formants; Press slides the mouth from "ah" to "ee".
Piano and Strings	Layer: the tine piano, and strings under it the wheel brings up.
Bass and Keys Split	Split at C3: an analog bass below, the reed piano above.

27 PRESETS

Flock

The creatures play the sound: from one species up to both, hunting each other, many through the mod buses.

Fireflies	Half Sine blips; a fast, scattered flock with short ink, spread wide.
Metal Rain	Inharmonic ×3.73 and ×7.11, scattered; each note detuned at random.
Starling Bells	Neighbour FM, Tune Pull and a cohesive flock; Nearest deepens the FM.
Dripping Ink	Long, wide ink; Turn, Speed and Heading play the filter and FM.
Ink Ribbons	Long ink and a loose lure; Turn plays cutoff and FM.
Murmuration Pad	Saws, an aligned flock spread wide; Heading pans each voice.
Pressure Front	Press opens 2:1 FM, with Neighbour FM.
Hunter Clicks	Layer: A flees, B hunts an octave up with Bite and Neighbour FM.
Mallet Flock	Layer: vibes and an upper shoal with Neighbour FM and Tune Pull.
Shark Bass	Split at C3: below, a hunting saw-and-pulse bass with Bite; Speed opens its filter.
Predator and Prey	Layer: A flees, B hunts an octave down; A's filter opens as its own kind closes in.
Murmuration Riot	Layer: a fast FM shoal opened by Press, Speed, Nearest and Turn, and driven ladder hunters that bite.
Murmuration Storm	Layer: a saw shoal with Neighbour FM and Tune Pull, and FM hunters.

Gathering Voices	A choir that swells and brightens as its creatures gather (Nearest, S-curved and slowed, on a bus), pulled into just intervals; Press lifts a creature, and a vibrato comes in only while one is pressed and climbing (LFO 1 × Press × Heading).
Hunting Grounds	Layer: A's glass plucks flee and ring brighter the nearer their pursuer; B's driven saws hunt and bite, their filter on the greater of Speed and Nearest.
Flight Steps	Arp Up as the amp, its steps quicker the faster each creature flies; Turn pans it, and a Global bus leans every note's timbre with the newest creature's heading.
Creature Theremin	Press lifts the creature up the sky; flying upward bends it sharp, diving flat, through a slow lag, and the wingbeat is its vibrato.
Scatter Glass	FM glass whose creatures keep apart: crowded, it dulls and quiets; alone, it opens up and shimmers.
Shoal Current	A shoal in a strong wind; how fast each creature is carried rings a comb tuned to its note, slowed into swells.
Circling Hawks	Layer: a low saw drone, and high sine hawks that hunt it and bite, only where a slow gust and their nearness agree (Min).
Crowd Swing	Gate 8 in time with the host as the amp; the more crowded a creature, the later its steps land (Warp), so a tight flock grooves.
Turning Strings	Strings whose bow roughens with how sharply each creature turns, either way (Absolute), with Tune Pull holding them together.
Two Flocks Talking	Layer: both parts ring each other's creatures; each part's chatter (Nearest × Speed on the FM) comes in a second or two into the chord.
Wingbeat Tremolo	LFO 1 × Speed on the level: a resting flock holds still, a flying one shivers.
Panic Bells	Bells fleeing a silent hunter (part B, heard only through its bite); the faster they run, the brighter and more out of tune.
Flock Organ	Detuned saws held in tune by Tune Pull; the flock's nearness and climb blend into one slow colour for the whole part (Global).
Ink Song	Long ink tails; each creature's speed and turning blend into a bus that sweeps a resonant band pass, and Press lifts its lure.

16 PRESETS

Pressure

Press shaped by the mod buses: curved, multiplied, delayed and lagged.

Lean-In Vibrato	A mono lead whose vibrato is LFO 1 × Press, fading in after a short delay; Press on an exponential curve is its breath.
Blown Reed	Press on an exponential curve is the breath: it speaks softly, then roars, with breath noise and a touch sharp on top.
Held-Back Bloom	Press slows Env 2's Bloom, and the filter follows the greater of the bloom and Press (Max).
Pressure Rotor	A drawbar organ whose rotor Press speeds up, on a lag over a second long like a rotor's inertia.

Touch Glass	Press on a Log curve, so a light touch already brightens the FM; the release velocity rings a shimmer as you let go.
Bowed Iron	Noise through a comb tuned to the note; the bow is the lesser of Press and Env 2's swell (Min).
Thumb on the Tape	Press sinks the pitch and darkens it, only near the top of its travel (Exponential), like a thumb on a tape reel.
Pressed Rhythm	A pad that Press turns into a rhythm: a synced Gate 8 × Press takes the level away in steps.
Restless Swarm	Press makes the creatures restless: faster, wandering and ringing their neighbours; how near they crowd sets the FM.
Patient Choir	Layer: hold a chord two seconds and a choir an octave up fades in over three more, as far as you press (a Global bus).
Squeezed Vowel	Press sweeps two band passes from "oo" toward "ah" on an S-curve, over a buzzing pulse.
Pressure Split	Split at middle C: Press drives the bass below, and brings a delayed vibrato to the lead above.
Finger Wah	Press sweeps a resonant low pass on an S-curve with a short lag, over a clavinet-ish pulse.
Holdable Pluck	A pluck that dies away, unless you press: Press holds it up, the harder the longer.
Pushing the Tide	A looped Surge breathes the filter; Press slows the tide and lifts it.
Dig-In Bass	Press raises the resonance and a sub; the drive bites only where Press and velocity agree (Min).

14 PRESETS

Expressive

Played with Press: it bows, blows, bends, gates or blooms.

Solo Violin	Press is the bow: it swells the note, brightens it and lets the vibrato in.
Breath Sax	Press is the breath: louder, brighter and, at the top, growling through the drive.
Accordion	Press is the bellows; two reeds a little apart, which the wheel pulls further.
Bending Lead	Press bends the note up a semitone, as a guitarist bends a string.
Pressure Growl	A bass that growls as you lean on it: Press drives the FM feedback, the drive and the resonance.
Bloom Bells	Strike a bell and it rings away; press and a pad grows out of it.
Tension Wire	Noise through a comb tuned to the note; Press tightens it toward a bowed metal ring.
Squeeze Organ	The tonewheel organ with Press on the rotor: lean in and it spins up, and the drive bites.
Wilting Pad	A pad that sags flat and darkens under pressure.
Flutter Pad	Press is the tremolo, deeper and faster the harder you press.
Pressure Metal	Press slides the FM modulator off its ratio, from a pure tone to a clanging bell.

Morph Pad	A low pass and a resonant high pass side by side: Press crossfades them, Slide moves the pulse width.
Press Gate Pad	A pad that only stutters when you lean on it: Press deepens a sixteenth-note gate.
Swell Brass	Brass that swells with Press: louder, brighter and a little sharper.

16 PRESETS

Drones

Shape envelopes and the mod buses keeping a held chord moving for minutes.

Basalt	Saws and a sub through a ladder; a Slow S fade-in, and Wander and Gusts shapes (23 s, 37 s) on the filter and pan.
Breathing Organ	Sine drawbars whose 2nd and 3rd partials breathe on their own Breath shapes, one reversed.
Cathedral Air	Noise in a comb tuned to the note; Shimmer on its ring, Twin Hills on the tone and the noise colour.
Undertow	A sub and a 1:2 FM growl that surges every 14 s; a Brownian walk turns its feedback.
Long Bloom	Chain 1 > 2: the amp blooms in over 12 s, and from 70% of the way the filter opens and closes on a 30 s arch.
Glacier	Chain 2 > 3 Loop: two 10 s arches start each other halfway, cross-fading a saw and FM glass forever.
Iron Hum	Saws in a comb with tube drive; Flicker on the comb, a 41 s Wander on the drive.
Tidal Pulse	Synced: four filter waves every eight bars, resonance wobbling down over sixteen.
Slow Ignition	Three seconds in, drive and FM fade up over eight more (a Global bus), so the whole chord catches fire; the wheel holds it back.
Singing Bowls	Noise through a comb tuned to the note, Shimmer moving its ring, and Tune Pull drawing the chord into just intervals.
One Sky	A Brownian walk sets the colour, width and pan of every note at once (a Global bus), so the chord drifts together.
Twin Pulse	Two slow LFOs multiplied: the drone swells only where both are up.
Rare Light	The filter follows the lesser of a 17 s Surge and a 23 s Plateau (Min), so it opens rarely and briefly.
Rust	Grit from the greater of a per-note random and a Flicker shape (Max), so each note crackles in its own way.
Far Horizon	Layer: high sines arrive four seconds into the chord and fade in over six (a Global bus) above a low fifth.
Drone Swarm	A vowel choir of creatures: Nearest and Heading, blended and slowed, set each voice's formants, Tune Pull holding the cluster.

17 PRESETS

Textures

Shapes as the amp, so each held note plays a pattern of its own, some paced by the flock.

Rain Glass	FM glass whose amp is the Sparkle shape, each note stretched by Random so a chord rains.
Bouncing Light	A dropped ball, over and over, faster up the keyboard; the filter bounces with it.
Echo Chamber	Each note echoes itself on the beat (Echoes, synced) and darkens over four bars.
Stutter Fields	Gates that speed up through each bar over a cutoff stepping at random.
Number Station	A band-passed tone keyed in sparse steps, with static, drifting in pitch.
Stairwell	The filter climbs a stair each beat and a fifth steps in at random.
Breathing Machine	Chain 1 > 2 > 3 Loop: the amp breathes, the filter strikes, the FM bounces, round again.
Night Pulse	A heartbeat, two beats a bar, under a 20 s wander through the dark.
Falling Bells	FM bells struck over and over by the Collapse shape, each tap lower, the FM fading with it.
Window Rain	Sparkle as the amp, its pace drawn per note; Press brings a downpour.
Gliding Weave	Random 16 in time with the host on Env 2, through a short lag so each step glides, on pitch, filter and pan.
Grain Swarm	Saw Train as the amp, its rate set by each creature's speed, its pitch smeared by how near its neighbour is.
Wind Chimes	A looping Bell as the amp in a hard wind; how fast the wind carries each creature sets how often it strikes.
Night Insects	High FM chirps on Stutter, the wingbeat bobbing each creature and its turning bending the chirp.
Clock Shop	Each note a clock on its own Heartbeat, its pace set by its key, its tick brightened by Press.
Tape Ghosts	A slow wow and a faster flutter summed into one bus, then multiplied by Press for a deeper wobble.
Ember Glow	Noise through a comb, crackling on Sparkle; the hottest embers, the greater of Press and a Gusts shape, ring brightest.

6 PRESETS

Motion

Patches that keep moving on their own.

Shifting Glass	Three-operator FM with Env 3 wandering on the FM (a 9 s Wander shape) and two unsynced LFOs; the wheel deepens all of it.
Random Garden	The Random wave: each note draws its own waves, LFO speed and pan.
Pulsar	A Pluck shape looping as the amp: every held note pulses on its own clock, faster higher up and with Press.
Clockwork	A sixteenth-note sample-and-hold on the FM and a dotted-eighth square on the filter.

MPE Voices	Two band passes for a vowel: Press brightens and adds vibrato, Slide moves the vowel, Release sets the tail.
Tide Machine	Layer: a saw drone breathing on slow LFOs, and FM bells an octave up echoing on an Echoes shape.

6 PRESETS

Interplay

Envelopes that move each other: one routed onto another's Time, Warp or Bend, or its own.

Tug of War	A filter arch that won't keep time: a slow wander onto its Time and Warp stretches and leans each pass.
Quickening	Gate 8 on the filter, its Time pulled down by a 40 s ramp, so slow pulses become a shiver.
Lingering Breath	A breath routed onto its own Time: it lingers when open and falls back quickly.
Mutual Orbit	Two shapes in orbit, each warping or bending the other, on FM colour and stereo place.
Tilting Steps	Random steps in time with the host, bent up and down by a slow arch.
Gathering Storm	Env 1 onto Env 2's Time: as the note blooms the flicker on the comb and noise gets faster.

12 PRESETS

Landscapes

Both parts at once, each on Shape envelopes that move each other: layers, a split, and Press, Slide and velocity crossfades. Some answer the macros.

Two Tides	Layer: saws open on a 13 s arch, FM glass swells on a 21 s one, each leaned by a wander. Macro 1 slows both, Macro 2 adds edge.
Ember Field	Layer: pink noise crackles in a tuned comb on Sparkle, paced by a wander, over a tube-driven sub that surges.
Glass Conversation	Layer: two bells on their own clocks, every 3 s and 4.7 s, a wander on each part speeding or slowing it. Macro 1 rings both faster.
Low and High Weather	Split at C3: surging saws bent by a random walk below, a looped chain of air and glass above.
Calm to Unrest	Press crossfade: a breathing sine choir, and a pulse swarm whose stutter quickens the harder you press.
Between Worlds	Slide crossfade: a looped chain of arches across saw and FM glass, and tempo-synced random steps.
Glass and Comets	Velocity: soft notes rain glass that settles as it ages, hard ones strike an 8 s comet.
Hunter's Drone	Layer: a pad, and B's hunters ring-modding it whenever one catches; the gusts make them fly faster.
Turning Choir	Layer: a dark pad under a choir whose root, fifth and octave hand over on a looped chain. Macro 1 slows it.
Clockwork Garden	Layer, in tempo: a heartbeat on B's amp, bent soft to hard by a 16-bar arch, under A's random filter steps.

Distant Engine	Layer: a driven comb hum, and B's Random wave spitting a saw train, slower on low notes.
Afterglow	Layer: a soft pad, and a fifth above that swells in late and lingers, its shimmer bent by the amp.

26 PRESETS

Monsta

Variations on MicroMonsta 2 programs (named in brackets), their aftertouch tricks on named Multiply buses.

Foggy EP	[Foggy EP] Soft keys behind a velocity-hungry filter; Press warbles one oscillator sharp, the wheel gives each note its own shape.
Genetic Lead	[Genetic] A plucky lead whose filter snaps shut; Press brings in a pulse two octaves up.
Monsta Voices	[MstaVoic] Formant voices that waver apart under Press; the wheel lets the vowel drift.
Fifty Pad	[Die-50] A resonant filter sweeping open and sinking over seconds, pulse widths shimmering; Press wobbles it.
Kingdome	[Kingdome] A driven three-oscillator pad; Press raises one oscillator up to a fifth and sets the filter wobbling.
Therapy Strings	[Therapy] Strings in reverse: Press hushes them while their shape shimmers.
Nightmare	[Nitemare] A low drone with a diving third oscillator; Press brings in its pulsing shriek.
Quack Bass	[MG Quak] A squelchy, driven mono bass with a long glide; Press sets the filter quacking.
Gristle Lead	[Gristle] Wide pulse waves whose widths swing; Press makes the resonance flutter.
Cumulus	[Cumulous] A wide pad under a resonant, tracking filter; Press speeds its drift, the wheel sweeps it across the stereo field.
Old Hills	[OldHills] Plucky keys over a held body; each note draws its own pulse width, and the pitch leans up as it settles.
Sunshine Strings	[Sunshine] Bright strings under a resonant, tracking filter, the width turning slowly; the wheel adds a fifth.
Worn Arp	[WornArp] Short plucks from a worn machine: each chirps, lands a little off pitch and somewhere new in the stereo field.
Waxxy	[Waxxy] A driven, resonant pluck whose filter lands somewhere new on each note.
Reptile	[Reptile] A crawling drone of three oscillators wandering on slow LFOs; Press drives it.
Crystal Strings	[Crystal] Octaves under a dark, resonant filter each note sweeps open and lets close over seconds.
Softy Drone	[Softy] A drone that takes seconds to arrive, its filter breathing; the wheel adds a fluttering resonance.
Ceasefire	[Ceasefre] A wide pad whose widths turn as it swells; Press brings in a pulsing fifth.
Chamber Strings	[ChamberS] Slow strings whose widths and levels move on an LFO; Press is the vibrato.
Fresh Air	[FreshAir] Wide saws under a filter that whistles at the note; Press sets the whistle wobbling.
Clang Heart	[ClangHrt] An inharmonic FM strike over a driven saw body; Press brings the clang back.

Space God	[SpaceGod] A howling, resonant swell rising over seconds; Press throws its top oscillator into a warble.
Opera Pad	[Opera2.0] A pad that sours as you press, one oscillator sinking flat; the wheel pulses in an octave.
Ice Bells	[IceBells] A filter resonant enough to ring, pinged by each note, brighter the harder you strike.
Oldfield Lead	[Oldfield] A second voice swells in under the lead as the vibrato speeds up.
Pink Haze	[PinkHaze] A hazy pad that glides between chords; Press pumps the filter, the wheel fades it.

> **INIT**
> *Init* is every parameter at its default: a saw through a low pass on part A, with the three starting routes (Env 2 and Press on the cutoff, Press on the flock's speed).

Your presets

User presets are small text files ending `.jspreset`, saved in:

macOS	`~/Library/Application Support/JSynth/Presets`
Windows	`%APPDATA%\JSynth\Presets`
Linux	`~/.config/JSynth/Presets (or $XDG_CONFIG_HOME/JSynth/Presets)`

Copy them between machines freely. A preset from a newer JSynth loads in an older one with any parameters it doesn't know skipped.

Favorites are kept in `Favorites.txt`, one preset name per line, in the folder above `Presets`. Every JSynth you open shares them, and they last between sessions.

CHAPTER 15

Sound Design Recipes

Starting points that show off what the flock can do. Each starts from Init unless it says otherwise.

DX electric piano

Two sines, a 1:1 pair for the body and a ×14 for the bark.

```
Osc 1 Sine · Osc 2 Sine Ratio ×1 Level 0
Osc 3 Sine Ratio ×14 Level 0
Env 1 A 1 ms D 5 s S 0 R 400 ms
Env 2 → FM 2 > 1 +30% · Env 3 (D 60 ms) → FM 3 > 1 +35%
Velocity → FM 2 > 1 +20% · Filter 1 open
```

Or load *Tine Piano*.

A flock that rings

Neighbours modulate each other, so a held chord shimmers as it gathers and breaks.

```
Osc 2 Sine Ratio ×3.5 · FM 2 > 1 40%
FLOCK FM 70% · TUNE PULL 40% · COHESION 70%
Nearest → FM 2 > 1 +30%
X PAN 60%
```

Or load *Starling Bells*.

Predator and prey

Two species; every catch rings, and the prey's filter opens as its own kind closes in.

```
Play mode LAYER
A: Flee · Nearest → F1 Cutoff +60%
B: Hunt · Speed 70% · Pulse · BITE 90% · Transp -12
```

Or load *Predator and Prey*.

Turning pad

The sound leans as each creature banks.

```
Osc 1 Alt Sine · FB 1 25%
TAIL 100% · SPREAD 90% · LURE 35%
Turn → F1 Cutoff +50% · Turn → FM 2 > 1 +40%
```

Or load *Ink Ribbons*.

Wind through a choir

A strong gust pushes the flock while LFO 2 sweeps the filter.

```
Saws, Noise 35% · F1 Band Pass, Res 40%
Wind Push 90% · Wind Rate 0.1 Hz
LFO 2 → F1 Cutoff +40% · Slide → F1 Res +30%
```

For a wind that plays the sound itself, load *Shoal Current*.

Pressure vibrato

Vibrato only where you press, arriving after the note settles.

```
LFO 1 5.5 Hz, DELAY 350 ms
Route LFO 1 → Fine at 0
Press → that route's amount +6%
```

As in *Pressure Lead*.

A stereo murmuration

The flock spreads a pad across the speakers as it turns.

```
Two saws, detuned · ALIGN 80% · SPEED 25%
X PAN 80% · DRIFT 15%
Heading → Pan +60%
```

Or load *Murmuration Pad*.

Struck drum

A pitch drop that goes further the harder you hit.

```
Osc 1 Sine · FB 1 15% · Env 1 D 300 ms S 0
Env 3 (D 45 ms) → Osc 1 Ratio +10%
Velocity → that route's amount +10%
```

APPENDIX A

Parameter Reference

Every parameter JSynth exposes to the host. Part parameters exist once for each part, named "A ..." and "B ..." in the host.

GLOBAL

PARAMETER	RANGE	DEFAULT
Play Mode	Single, Layer, Split, MPE Zones, Velocity, Expression	Single
Split Key	C-1 – G9	C4
Velocity Split	1 – 127	80
Crossfade Source	Slide, Press	Slide
Quality	1x, 2x, 4x	2x
Master Level	-60 – +6 dB	-6.0 dB
Output Width	0 – 200%	100%
Manager Bend Range	1 – 96 st	2 st
Glide Rounding	0 – 100%	0%
Rounding Time	0 ms – 20 s	150 ms
Macro 1 – 4	0 – 100%	0%
Velocity ... Release Curve	-1 – +1	+0.00
Velocity ... Release Smoothing	0 ms – 20 s	8.0 ms (Glide 2.0 ms)
Size	0 – 100%	50%
Walls	Soft, Wrap, Open	Soft

VOICE (PER PART)

PARAMETER	RANGE	DEFAULT	
Level	-60 – +6 dB	0.0 dB	ROUTE
Pan	-100% – +100%	+0%	ROUTE
X Pan	0 – 100%	0%	ROUTE
Transpose	-48 – +48 st	+0 st (B: -12 st)	
Fine	-100 – +100 ct	+0 ct	ROUTE
Key Low, Key High	C-1 – G9	C-1, G9	

PARAMETER	RANGE	DEFAULT	
Polyphony	1 – 16	8	
Voice Mode	Poly, Mono, Legato	Poly	
Glide	0 ms – 20 s	0.0 ms	ROUTE
Glide Mode	Always, Legato	Always	
MPE Bend Range	1 – 96 st	48 st	
Drift	0 – 100%	0%	ROUTE
Velocity	0 – 100%	100%	
Output	Main, Own	Main	

OSCILLATORS (PER PART: OSC 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Wave	Sine, Triangle, Saw, Square, Pulse, Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel, Alt Camel, White, Pink, Random	Saw (Osc 2, 3: Sine)	
Shape	0 – 100%	50%	ROUTE
Pitch Mode	Note, Ratio, Fixed	Note (Osc 2, 3: Ratio)	
Coarse	−48 – +48 st	+0 st	ROUTE
Fine	−100 – +100 ct	+0 ct	ROUTE
Ratio	×0.25 – ×16	×1 (Osc 3: ×3)	ROUTE
Fixed	1 Hz – 10 kHz	440 Hz	ROUTE
Sync	Off, On	Off	
Retrigger	Off, On	On	
Phase	0 – 360°	0°	ROUTE
Bend	0 – 100%	100%	
Key Track	Off, On	On	
Filter	1+2, F1, F2	1+2	
Level (mixer)	0 – 100%	100% (Osc 2, 3: 0%)	ROUTE
Pan (mixer)	−100% – +100%	+0%	ROUTE

FM, NOISE AND DRIVE (PER PART)

PARAMETER	RANGE	DEFAULT	
FM Feedback 1 – 3, FM 1 > 2 … 3 > 2	0 – 100%	0%	ROUTE
FM Type	Phase, Exponential	Phase	
Noise Level	0 – 100%	0%	ROUTE

PARAMETER	RANGE	DEFAULT	
Noise Color	-100% – +100%	+0%	ROUTE
Noise Filter	1+2, F1, F2	1+2	
Drive Type	Soft, Hard, Tube	Soft	
Drive	0 – 100%	0%	ROUTE

FILTERS (PER PART: FILTER 1, 2)

PARAMETER	RANGE	DEFAULT	
Filter Routing	Serial, Parallel	Serial	
Filter Balance	-100% – +100%	+0%	ROUTE
Type	Low Pass, High Pass, Band Pass, Notch, Peak, Ladder, Comb +, Comb –	Low Pass (F2: Ladder)	
Slope	12, 24, 36, 48 dB	24 dB	
Cutoff	20 Hz – 20 kHz	1.26 kHz (F2: 20.00 kHz)	ROUTE
Resonance	0 – 100%	20% (F2: 0%)	ROUTE
Drive	0 – 100%	0%	ROUTE
Key Track	0 – 100%	0%	ROUTE

ENVELOPES (PER PART: ENV 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Delay	0 ms – 20 s	0.0 ms	ROUTE
Attack	0 ms – 20 s	5.0 ms	ROUTE
Hold	0 ms – 20 s	0.0 ms	ROUTE
Decay	0 ms – 20 s	400 ms (Env 2, 3: 300 ms)	ROUTE
Sustain	0 – 100%	70% (Env 2, 3: 30%)	ROUTE
Release	0 ms – 20 s	300 ms	ROUTE
Attack, Decay, Release Curve	-1 – +1	+0.00	
Loop	Off, Held, Free	Off	
Velocity	0 – 100%	0%	
Time Range	Short, Normal, Long	Normal	
Mode	ADSR, Shape	ADSR	
Shape	60 shapes in 10 banks	Swell: Arch	
Shape Time	1 ms – 30 min	2.00 s	ROUTE
Shape Sync	Off, On	Off	

PARAMETER	RANGE	DEFAULT	
Shape Length	1/16 – 64 bars	1 Bar	
Warp	-100% – +100%	+0%	ROUTE
Bend	-100% – +100%	+0%	ROUTE
Repeat	Once, 2x – 16x, Held, Forever	Once	
Reverse	Off, On	Off	
Retrigger	Smooth, Restart, Ignore	Smooth	
Chain Phase	0 – 100%	100%	ROUTE
Sustain Point	0 – 100%, Off	Off	ROUTE
Shape Level	0 – 100%	100%	ROUTE
Env Chain (per part)	Off, 1 > 2, 2 > 3, 1 > 2 > 3, and each as a loop	Off	

LFOS (PER PART: WINGBEAT, WIND, LFO 1, LFO 2)

PARAMETER	RANGE	DEFAULT	
Shape	Sine, Triangle, Saw, Square, S&H, Smooth	Sine (Wind: Smooth, LFO 2: Triangle)	
Rate	0.02 – 50 Hz	5.00, 0.20, 4.00, 0.50 Hz	ROUTE
Sync	Off, On	Off	
Division	1/32 – 4/1	1/4	
Trigger	Retrigger, Free, Global	Retrigger (Wind: Global)	
Phase	0 – 360°	0°	ROUTE
Delay	0 ms – 20 s	0.0 ms	ROUTE
Polarity	Bipolar, Unipolar	Bipolar	
Push (Wingbeat, Wind)	0 – 100%	20%, 30%	ROUTE

BUSES (PER PART: BUS 1 – 4)

PARAMETER	RANGE	DEFAULT	
Base	-100% – +100%	+0%	ROUTE
Mode	Sum, Multiply, Min, Max	Sum	
Curve	Linear, Exponential, Log, S-Curve, Absolute	Linear	
Polarity	As Is, Unipolar, Bipolar, Invert	As Is	
Delay	0 ms – 20 s	0.0 ms	
Fade	0 ms – 20 s	0.0 ms	
Lag	0 ms – 20 s	0.0 ms	
Global	Off, On	Off	

FLOCK (PER PART)

PARAMETER	RANGE	DEFAULT	
Behaviour	Shoal, Scatter, Hunt, Flee	Shoal (B: Hunt)	
Cohesion	0 – 100%	50%	ROUTE
Alignment	0 – 100%	50%	ROUTE
Separation	0 – 100%	40%	ROUTE
Speed	0 – 100%	40% (B: 55%)	ROUTE
Wander	0 – 100%	20%	ROUTE
Lure	0 – 100%	60%	ROUTE
Range	0 – 100%	35%	ROUTE
Neighbour FM	0 – 100%	0%	ROUTE
Tune Pull	0 – 100%	0%	ROUTE
Bite	0 – 100%	0%	ROUTE
Tail Length	0 – 100%	50%	ROUTE
Tail Spread	0 – 100%	50%	ROUTE
Lure X, Lure Y	Pitch, Velocity, Press, Slide, Release, Random, Centre	Pitch, Slide	

ROUTES (PER PART: ROUTE 1 – 64)

PARAMETER	RANGE	DEFAULT	
Source	Off and 25 sources	Off (1–3: Env 2, Press, Press)	
Destination	181 destinations	Filter 1 Cutoff (2: Speed)	
Amount	−100% – +100%	+0% (1–3: +30, +40, +20%)	ROUTE

APPENDIX B

Specifications

Format	CLAP 1.2 instrument. Features: instrument, synthesizer, stereo.
Ports	One note input (CLAP, MIDI and MIDI MPE; CLAP preferred). Three stereo outputs: Main, Part A, Part B.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit).
Latency	None.
Parts	Two, each a complete patch and a species in the flock.
Voices	Up to 16 per part (40 slots in all, so stolen notes fade out in 3 ms). Mono and Legato with a note stack.
Oscillators	3 per voice, 15 waves (Random among them), double-precision phase. Saw, square and pulse band-limited by polyBLEP.
FM	3 × 3 grid including feedback; phase or exponential. Plus Neighbour FM from up to three neighbours.
Filters	2 per voice: TPT state variable (12–48 dB), notch, peak, 4-pole ZDF ladder, comb ±. Serial or parallel, each oscillator and the noise picking its filter.
Modulation	3 envelopes, each a DAHDSR with curves and loop or a Shape (60 contours, 1 ms to 30 min, chainable); 4 LFOs (2 move the flock); 4 mod buses; 5 touch dimensions, 4 flock sources, key, random, mod wheel, 4 macros. 64 routes per part.
Oversampling	1×, 2× (default) or 4×, decimated by polyphase IIR half-bands.
Control rate	Every 32 samples, ramped per sample. The flock steps at 500 Hz. Renders are identical at any buffer size.
Parameters	All automatable except the route sources, destinations and Quality. Continuous ones accept CLAP modulation; route destinations accept it per note (note id, key, channel or port).
State	Saved with the host project, including the preset name. States from older and newer builds load, with unknown parameters skipped and missing ones at their defaults.
Panel	1648 × 940 at 100%; 75%, 100%, 125%, 150% and 200%.
Threading	The audio thread never allocates, locks or waits on the panel after activation.

Troubleshooting

SYMPTOM	WHAT TO CHECK
JSynth doesn't appear in my host (macOS)	The download is probably quarantined. See the warning in Chapter 2, then rescan plug-ins.

SYMPTOM	WHAT TO CHECK
Only part A plays	The play mode is **SINGLE**. Choose **LAYER** or another mode in the header. Check each part's Key Low and Key High too.
Turning up the flock knobs changes nothing in the sound	By design: the flock only reaches the sound through routes and the three BETWEEN CREATURES knobs. Map a flock source, or raise **FLOCK FM**.
FLOCK FM makes no difference	It modulates Osc 2. Give Osc 2 some **LEVEL**, or route it into Osc 1 with **2 > 1**. Creatures also have to be near each other: raise **RANGE** or **COHESION**, and play more than one note.
BITE does nothing	Bite is set on the hunting part, and needs the hunter's Osc 1 and a prey of the other species: play mode LAYER (or any mode that plays both), one part on **Hunt**.
Notes are out of tune	Check **TUNE PULL** (just intervals are up to 30 cents from equal temperament), **DRIFT**, and a route onto Fine or Ratio. An MPE controller sends bend per note at the part's **BEND** range, which must match the controller's (48 semitones is the MPE standard).
Pitch bend is far too wide on a plain keyboard	A plain keyboard is on channel 1, which bends by **MGR BEND** (2 semitones). If its notes come on other channels, they are taken as MPE member channels and bend by **BEND**.
Notes cut off when I play a lot	The part is at its polyphony and the oldest notes are being taken. Raise the voices count on the VOICE tab (up to 16).
My user presets are missing	Check the preset folder for your OS (Chapter 14). Files must end in `.jspreset`.

Glossary

Creature A note, as it flies in the sky. Each sounding voice has one.

Species A part. Each part's creatures flock with their own kind.

Lure Where a held note pulls its creature.

Boids The classic flocking model (Craig Reynolds, 1986): each creature steers only by its neighbours, by cohesion, alignment and separation. See Chapter 11.

Neighbour A creature of the same species within RANGE. Only neighbours flock together and hear each other.

Hunter, prey A creature of a part on Hunt, and the creature of the other species it chases. A catch turns on Bite.

MPE MIDI Polyphonic Expression: each note on its own MIDI channel, so each can bend and be pressed on its own.

FM Frequency modulation: one oscillator wobbling another's phase or pitch at audio rate, for bright, bell-like and metallic tones.

Ratio An oscillator's frequency as a multiple of the note's.

Just interval An interval from a simple whole-number ratio, such as 3:2 for a fifth. They beat less than equal-tempered ones.

Route A source, a destination and an amount.

Ring modulation Multiplying two signals, for clangorous sum-and-difference tones.



JSynth

THE FLOCKING SYNTH · V0.10.4

github.com/jeffmhopkins/Jclap